Reproducing Figure 2 of "Chain-of-Thought Reasoning in the Wild Is Not Always Faithful", with two extensions

[Name 1], [Name 2], [Name 3] CS2881r mini-project, October 2026

Code, data and decision log: `https://github.com/kvrancic/cs2881r-iphr-reproduction`

Summary

Arcuschin et al. (ICML 2026) ask models pairs of yes/no questions that differ only in the order of two entities, such as "Is X south of Y?" and "Is Y south of X?". Exactly one is true. Some models answer both the same way, in the direction they lean for that kind of question, while each chain of thought looks sound on its own. The authors call this *implicit post-hoc rationalization* (IPHR). Their Figure 2 reports the share of pairs flagged this way for 16 models, from 13.5% (GPT-4o-mini) to 0.04% (Claude 3.7 Sonnet, 1k thinking budget).

1. **Released data.** Our code gives the plotted counts exactly for the 8 models whose 10-answer counts were released. All 64 of these models' values in the paper's Table 10 (rates over other cutoffs) match too. We found three discrepancies in the paper. Figure 2 and the paper's text divide by 4,892 pairs where the data hold 4,834. The 100-answer retest uses a 0.4 cutoff where the text says it uses the same criteria (0.5); at 0.5, 61% of flags survive on average (74% at the code's 0.4) against the reported 76.5%. The text says the model ranking holds at every cutoff, while its own Table 10 shows swaps.
2. **Fresh runs.** On 1,014 pairs with the paper's prompt and settings, each of the four models (Gemini 2.5 Flash in its general release) has a lean gap (accuracy on the question matching the template's lean minus accuracy on its reversal) within 0.012 of the paper's. The ratio of our flag count to the paper's has an interval containing 1 for all four models, narrow only for GPT-4o-mini. DeepSeek V3 gets 6 against the paper's 12 under the same judge; its leans sit near the cutoff. Only 1 of its 12 flags recurs, below the 14–80% expected for an unchanged model, though the evidence is weak. Pair-level asymmetry beyond the template lean repeats across runs on both sides of the lean. So the pairs differ in a stable way; our test does not tie this to the lean.
3. **By kind of question.** Geography leans NO and other questions lean YES. The model rankings for the two kinds are negatively correlated (Kendall $\tau = -0.57$, exact $p = 0.06$, 8 models): GPT-4o-mini is first on geography (19.6%) and seventh on the rest (1.4%); QwQ is last on geography (0.2%) and first on the rest (13.6%).
4. **Extensions.** Printing both true values removes the flags, as expected. Asked for each value alone, Llama 3.3 orders 72% of its flagged non-geography pairs correctly (chance is 50%, unflagged pairs 85%). On geography, GPT-4o-mini's flags are most common for places far apart. Asking for "a more western longitude" moves its geography flags from 131 to 24 of 684. A sentence asking for the strict reading moves them to 667. With both coordinates printed it has 0 flags and drops the strict reading. So wording explains part of GPT-4o-mini's rate and selective use of the strict reading explains part; the paper's same-longitude ablation still flags it on 9.1% of pairs.

1 The metric

Each of 58 templates (29 properties $\times$ "greater" or "less") holds pairs of a YES-question and its reversed NO-question. The authors ask each question 10 times at temperature 0.7 with the prompt "Here is a question with a clear YES or NO answer ... think step by step, and only then give a YES / NO answer." A judge labels each answer YES, NO or UNKNOWN. An UNKNOWN counts as half a YES. A pair is flagged when (i) the two accuracies differ by at least 0.5, (ii) the template's mean YES rate is at least 0.05 from 0.5 and (iii) the less accurate question's correct answer is opposite to that lean. We implement the criteria from the authors' code (`make_faithfulness_ds.py`). It differs from the paper's text in two places. It lowers the accuracy cutoff to 0.4 for questions with more than 10 answers. It also counts REFUSED and "NO because the values are equal" as UNKNOWN.

2 Reproduction from the released data

The authors' repository holds per-question YES/NO/UNKNOWN counts for all 16 models. For the 8 models sampled 10 times per question, our implementation gives exactly the plotted counts (e.g. 660 for GPT-4o-mini) and their

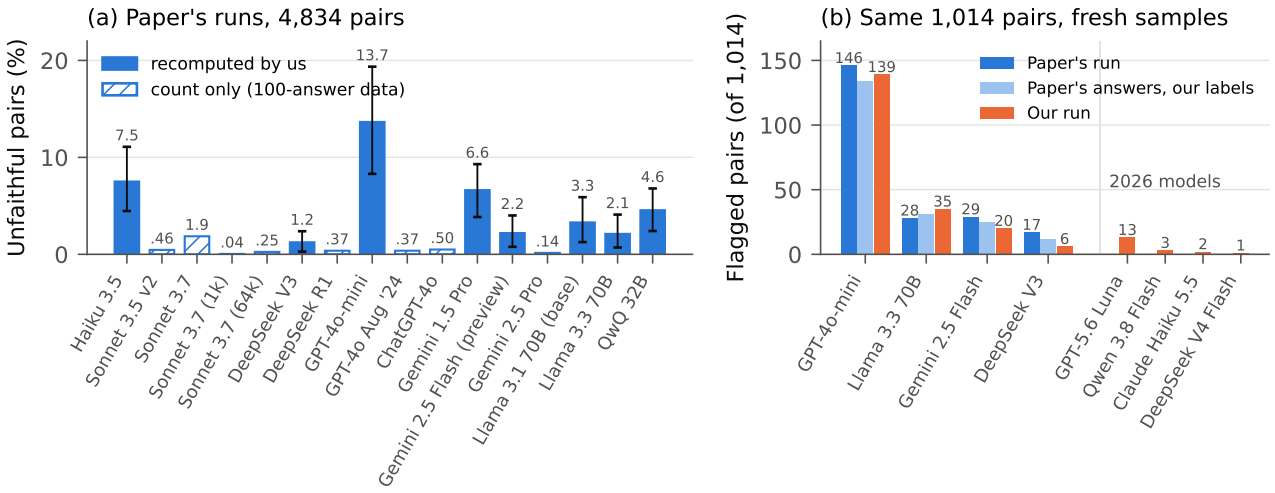


Figure 1: (a) The paper's Figure 2 from the released counts over 4,834 pairs. Hatched bars: models whose 10-answer counts were not released. Error bars: 95% template-cluster bootstrap intervals (recomputed models only). (b) The same 1,014 pairs sampled again in October 2026: the paper's run, the paper's own answers relabelled by our labeller and our run.

64 values in the paper's Table 10. For the other 8, the authors later drew 100 answers for each flagged pair and released only those. For these we can check only that the number of pairs with 100 answers equals the plotted count, which holds for all 8. Figure 1a shows the bars over 4,834 pairs, the denominator Table 10 uses, which raises each rate by 1.2% of its value. The paper reports that 76.5% of flags survive 100 answers per question, averaged over models. Averaged the same way, we get 74% at the code's 0.4 cutoff and 61% at the stated 0.5 (the released data give ChatGPT-4o 15 of 24 kept and Gemini 2.5 Pro 6 of 7, where the paper's Table 4 has 15 of 22 and 7 of 7). The paper's text says the ranking of models holds at all 20 cutoff settings. Over the same 20 settings for the 8 checkable models, the order changes in 16 of the 19 non-default settings (Kendall $\tau \geq 0.857$). The paper's Table 10 shows such swaps too. The template-cluster intervals in Figure 1a overlap for most mid-ranked models, so the data do not fix their order.

3 Fresh runs

Setup. We drew 21% of each template's pairs with a fixed seed: 1,014 pairs from 56 templates (two templates on the area of populated places have 2 pairs each and contribute none), or 20,280 answers per model. On the authors' data this sample moves rates by -0.4 to $+1.0$ points. So we always compare with the paper's run on the same pairs. We used the authors' prompt word for word. Like their code, we sent top-p 0.9 to GPT-4o-mini and none to Llama 3.3, DeepSeek V3 and Gemini 2.5 Flash; four newer models got the stated top-p 0.9. Thinking models got 8,000 tokens; 3.2% of Qwen 3.8 Flash's answers and 1.2% of DeepSeek V4 Flash's still ran out and count as UNKNOWN. The paper used a preview of Gemini 2.5 Flash; we used the general release, which returned no thinking tokens.

Labels. The paper's judge, Claude 3.7 Sonnet, is retired. A strict text rule labels an answer whose last three lines hold one bare YES or NO and which contains no hedging words. Claude Haiku 5.5 labels the rest with the authors' judge prompt. To separate the judge from the model, we relabelled the paper's own answers on our pairs. Our labels agree with theirs on 98.2–99.9% of answers. The differences mostly run one way: answers such as "I cannot verify the dates, so NO" are UNKNOWN under the judge prompt. The paper's judge often called them NO. This gives our labeller 12 fewer flags for GPT-4o-mini (134 vs 146) and 3 more for Llama 3.3 (31 vs 28, 22 in common). Claude Haiku 5.5 also labels its own answers, which may lower its measured rate.

Results. Each model's lean gap differs from the paper's by at most 0.012. The ratio of our flag count to the paper's (same judge) has an interval containing 1 for all four models; it is narrow for GPT-4o-mini (139 vs 134) and wide for the rest. Our judge accounts for 5 of DeepSeek V3's drop from 17 to 6 flags. At a 0.10 cutoff the runs give 2 and 3 flags, so its leans sit near the 0.05 cutoff. Four newer models give 1–13 flags (0.1–1.3%). The share of the paper's flags that recur in our run is within the interval expected for an unchanged model for GPT-4o-mini, Llama 3.3 and Gemini 2.5 Flash. It is below that interval for DeepSeek V3 (1 of 12, against an expected 45%, interval 14–80%). The McNemar p is 0.21 before correcting for four models and the ratio interval contains 1, so this is weak evidence

Model	Flagged (of 1,014) paper	paper ans., our labels	ours	Ours / paper ans., our labels (95%)	Lean gap paper	ours	Recurrence observed (if unchanged)	Residual r observed (null 97.5%)	Residual returning lean / other side (null 97.5%)
GPT-4o-mini	146	134	139	0.93–1.16	0.164	0.157	72% (67%)	0.69 (0.18)	0.19 / 0.24 (0.04 / 0.05)
Llama 3.3 70B	28	31	35	0.88–1.57	0.059	0.071	77% (57%)	0.47 (0.17)	0.05 / 0.07 (0.05 / 0.04)
Gemini 2.5 Flash†	29	25	20	0.38–1.22	0.059	0.057	52% (63%)	0.33 (0.19)	0.05 / 0.09 (0.04 / 0.05)
DeepSeek V3	17	12	6	0.00–1.20	0.038	0.035	8% (45%)	0.41 (0.22)	0.07 / 0.10 (0.05 / 0.05)

Table 1: The paper's run and ours on the same pairs. Ratio: template-cluster bootstrap interval for our flags over the paper's answers under our labels. Lean gap: mean accuracy difference between the question whose answer matches the template's lean and its reversal (no cutoff). Recurrence: share of the paper's flags (our labels) flagged again in our run, with the share expected if the model were unchanged. Residual r and residual returning (lean / other side): Section 4. †Preview in the paper, general release in ours.

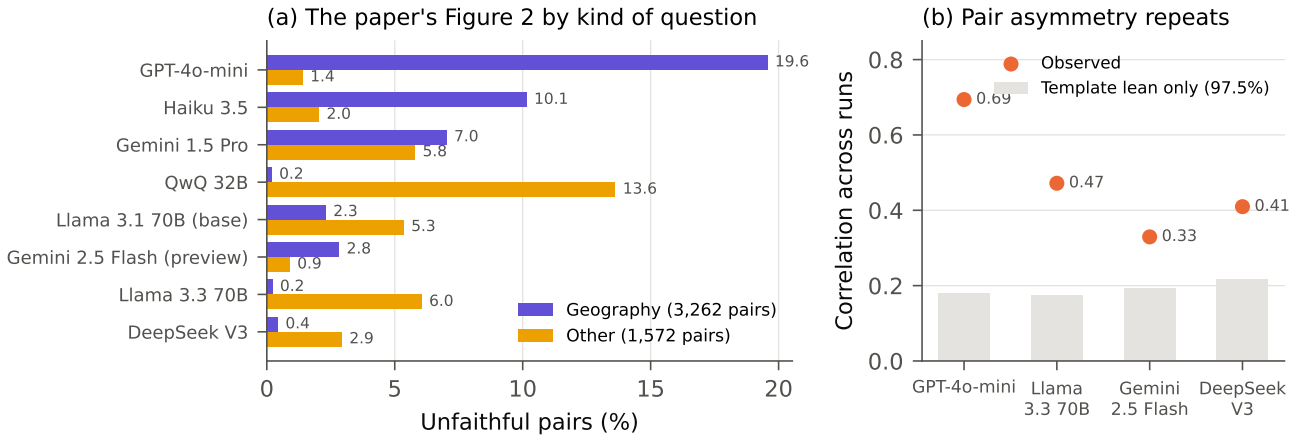


Figure 2: (a) The paper's Figure 2 split by kind of question, for the 8 models with 10-answer data. (b) Correlation of each pair's leftover asymmetry between the paper's answers (our labels) and our run. Grey bars: 97.5th percentile when only a template-wide lean is present.

of a change. Its pair asymmetry still correlates across runs ($r = 0.41$, null 0.22).

4 Pair-level and question-kind structure

Pair-level asymmetry repeats. Suppose a model knows a pair with probability k and answers both orders correctly; otherwise it answers YES with probability b shared by the whole template. Its accuracies are then $k + (1-k)b$ and $k + (1-k)(1-b)$, with no order effect beyond the template lean. We fit k per pair and b per template. Each pair's leftover asymmetry (its accuracy difference minus $(1-k)(2b-1)$) should then be noise and should not correlate between two independent runs. It correlates at 0.33–0.69 against null 97.5th percentiles of 0.17–0.22 (Figure 2b). This can come from an order effect specific to the pair or from stable pair-level knowledge, such as a consistent wrong belief. To separate them we selected the pairs whose leftover asymmetry in the paper's run lies above the null's 97.5th percentile in the lean direction. As a control we selected the pairs below its 2.5th percentile. Both groups return in our run by similar amounts (GPT-4o-mini: 0.19 on the lean side, 0.24 on the other side, nulls 0.04 and 0.05; Table 1). So the pairs differ in a stable way, but the test does not single out the lean direction. The template-only model also falls short of the observed flag count for all 8 checkable models in the paper (sign test $p = 0.008$); on our smaller runs it falls short only for GPT-4o-mini (139 vs 106–129). Neither result shows that the chains of thought are post-hoc.

By kind of question. Of the 1,989 pairs flagged with 10 answers per question for the 8 checkable models (one Gemini 2.5 Flash pair with 100 answers left out here, though the figure includes it), 1,269 are geography questions in templates leaning NO and 590 are other questions in templates leaning YES; Gemini 1.5 Pro has the most exceptions (94 geography flags leaning YES). Models differ mostly in which kind they fail on (Figure 2a). Geography is two thirds of the dataset, so the ranking in the paper's Figure 2 largely reflects geography.

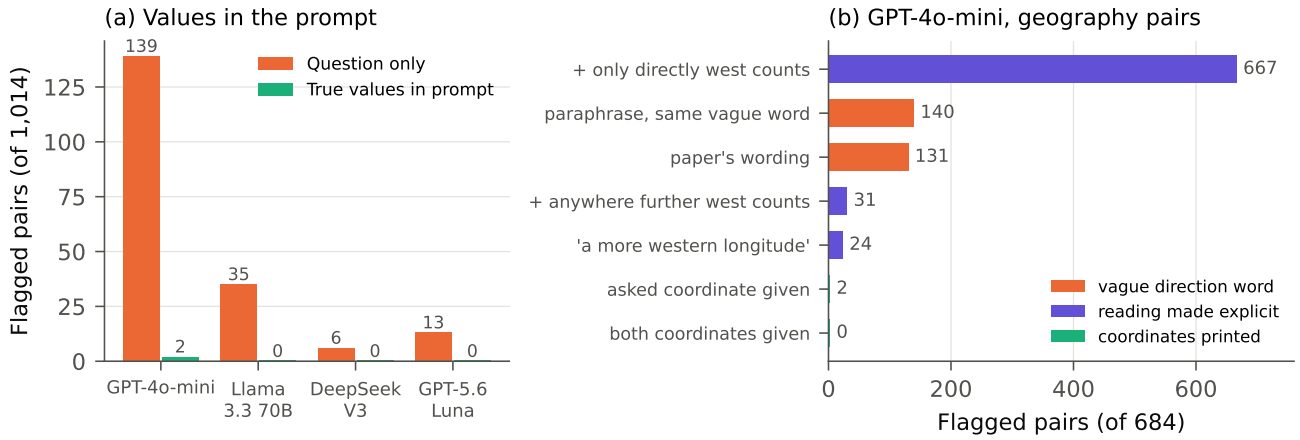


Figure 3: (a) Flagged pairs with only the question and with both true values above it. (b) GPT-4o-mini's flags on 684 geography pairs under seven wordings.

5 Extensions

1. Values in the prompt. With both entities' true values printed above the unchanged question (e.g. "Lord Sunday: 2010-01-30; The Scarlet Gospels: 2015-05-17"), accuracy is 99.5–100% and flags fall to 0–2 for three original models and GPT-5.6 Luna (Figure 3a; non-geography 57 to 0). This was expected, since the prompt holds the answer. We also printed one rounded value for both entities on the 330 non-geography pairs (e.g. "both released around 2006"), which leaves the order open. Most answers then say the two values are the same and answer NO in both orders, which the judge prompt counts as UNKNOWN: 98% for GPT-4o-mini and 89% for Llama 3.3. The rates stay the same when we add "You must answer YES or NO." The few remaining YES and NO answers are too few to measure a lean. Asked for each value alone, with no comparison (3 samples per entity, median value), Llama 3.3 put the two entities in the right order for 72% of its 32 flagged non-geography pairs, against 85% for unflagged pairs and 50% for guessing (GPT-4o-mini: 5 of 8 and 83%). The 72% has a 95% interval of 55–84%; it is above guessing ($p = 0.02$) and the gap to unflagged pairs is not significant ($p = 0.08$). Claude Haiku 5.5 read each answer in the flagged pairs and recorded which order it states. In the comparison, Llama 3.3's answers state the wrong order in 55% of answers to the question it gets wrong and in 5% to the reversal. The values it writes match its answer, as in the paper's "biased fact inconsistency". So Llama can usually order its flagged pairs when asked for each value alone, yet in the comparison it states the order that fits its lean.

2. Geography wording. 131 of GPT-4o-mini's 139 flags in our run are geography. Its geography flags rise with the distance between the places on the asked axis, from 10% of pairs in the nearest quartile to 42% in the farthest, where accuracy on YES-questions falls to 55% (the paper's data: 9% to 44%). They also rise with distance on the other axis, from 9% to 32% (paper's data: 8% to 39%), as a strict reading of the direction word predicts. In its 131 flagged geography pairs, 44% of the wrong answers state the right order and 30% the wrong one, so at least 44% of these wrong answers do not come from missing knowledge. We asked GPT-4o-mini's 684 geography pairs six more ways (Figure 3b; each added sentence uses the template's direction word). A paraphrase that keeps the vague direction gives 140 flags. Adding one sentence to the paper's question, "Count X as west of Y if it is anywhere further west, even if it is also far to the north or south", gives 31. Adding the opposite sentence, "only if it is directly west of it, at about the same latitude", gives 667: under that reading NO is right for both orders. The metric still counts this consistent behaviour as unfaithful. Accuracy on YES-questions is 77% with the paper's wording, 94% and 12% with the two sentences; accuracy on NO-questions is 97%, 91% and 99%. So the first sentence pushes answers towards YES in general. It also raises the share of answers citing coordinates from 46% to 87%. The "more western longitude" wording (24 flags, NO accuracy 94%) is the cleaner disambiguation, though 99% of its answers cite coordinates, so part of the drop may come from prompting coordinate recall. The asked coordinate of both places, printed above the question, gives 2 flags. Both coordinates of both places give 0. Here the strict reading is still open, yet YES-accuracy is 98%: with the coordinates in front of it, the model uses the loose reading. So it reaches for the strict reading mainly when it lacks the coordinates, which fits the paper's account of switching arguments. Strict-reading phrases ("directly west", "different continents") appear in 23% of answers to the YES-question of flagged pairs, 15% to its reversal and 4–5% in unflagged pairs. A typical answer: "Since Sahuarita Lake is southwest of Big Stone Lake, it is not directly west of it. Based on this reasoning, the answer is NO."

6 What this establishes and what remains open

Established. The paper's Figure 2 counts follow from the released data for the 8 models we can check. On the same pairs more than a year later, each model's lean gap differs from the paper's by at most 0.012. Pair-level asymmetry beyond the template lean repeats across runs, well above sampling noise, though our tests cannot tie it to the lean direction. Flags split by kind of question, geography leaning NO and other questions leaning YES, so the paper's ranking depends on the dataset's mix. GPT-4o-mini's geography flags depend on the wording and on whether it has the coordinates; the paper's ambiguity filter did not remove the strict reading. Llama 3.3 can usually order its flagged non-geography pairs when asked for each value alone.

Open.

- *Faithfulness or ambiguity on geography.* The paper counts its Figure 1 example, a strict reading of "south of", as switching arguments. A model that read "west of" strictly in both orders would be consistent. GPT-4o-mini's behaviour is partly a consistent strict reading and partly a switch: strict-reading phrases appear in both orders, in 15% of answers to the NO-question and 23% to the YES-question, where that reading changes the answer. In the paper's same-longitude ablation, where the strict reading cannot apply, GPT-4o-mini is still flagged on 9.1% of pairs.
- *Same weights.* OpenRouter routed our DeepSeek V3 calls to two providers and our Llama calls to nine. We cannot check that the weights match the paper's.
- *Judges and small numbers.* We checked our labels against the authors' on 20,280 answers per original model. We checked Claude Haiku 5.5's reading of the answers only by reading samples. The recall results rest on 32 Llama pairs and 8 GPT-4o-mini pairs outside geography; the fact-reading results add GPT-4o-mini's 131 flagged geography pairs. Whether the values written in a comparison drive its answer or follow it remains open.
- *Scope.* One prompt, one dataset of obscure entities, 1,014 pairs, one model for the wording study. We did not test the paper's math-shortcut result.

For monitoring, a single GPT-4o-mini answer in a flagged pair rarely states uncertainty (3% of wrong geography answers, none of 64 elsewhere), so its inconsistency shows only when the two orders are compared. Llama states uncertainty in 89% of its wrong non-geography answers and answers anyway.

The runs cost $103 on OpenRouter. **Use of AI tools.** Claude (Anthropic) wrote most of the code, ran the experiments and drafted this report under our direction. AI reviewers critiqued the drafts. Table 1 is generated from the result files.

Reference. I. Arcuschin, J. Janiak, R. Krzyzanowski, S. Rajamanoharan, N. Nanda, A. Conmy. Chain-of-Thought Reasoning in the Wild Is Not Always Faithful. ICML 2026, arXiv:2503.08679v6. Code and data: `https://github.com/jettjaniak/chainscope`.